

B2 Symmetry & Relativity — Model Solutions, 2018

Question 1 — 4-vectors, kinematics, dynamics

(a) Intervals, null separation, orthogonality

Problem. Conditions for spacelike/timelike intervals between events $D = (ct_d, \mathbf{x}_d)$, $G = (ct_g, \mathbf{x}_g)$. Null separation? Conditions $Y^\mu X_\mu = 0$. Which pairs of A, K, J are orthogonal?

Define the interval 4-vector:

$$\Delta X^\mu = D^\mu - G^\mu = (c\Delta t, \Delta \mathbf{x}).$$

Identify the components:

$$\Delta t = t_d - t_g, \quad \Delta \mathbf{x} = \mathbf{x}_d - \mathbf{x}_g.$$

Form the Minkowski-square in signature $(+, -, -, -)$:

$$\Delta X^\mu \Delta X_\mu = c^2 \Delta t^2 - |\Delta \mathbf{x}|^2.$$

State the timelike condition (causally connected by sub-light signal):

$$c^2 \Delta t^2 - |\Delta \mathbf{x}|^2 > 0$$

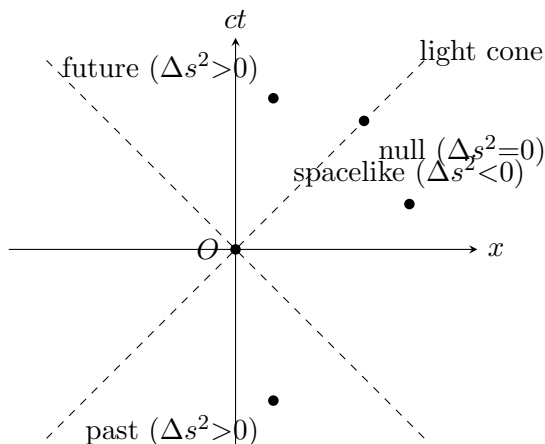
State the spacelike condition (no causal signal possible):

$$c^2 \Delta t^2 - |\Delta \mathbf{x}|^2 < 0$$

State the null (light-like) condition:

$$c^2 \Delta t^2 - |\Delta \mathbf{x}|^2 = 0$$

Yes, two events can be connected by a null vector — a photon emitted at G and absorbed at D traces such a null worldline.



When is $Y^\mu X_\mu = 0$? Three independent cases:

1. One of the 4-vectors vanishes: $Y^\mu = 0$ or $X^\mu = 0$.
2. Both are null and parallel: $Y^\mu Y_\mu = X^\mu X_\mu = 0$ with $Y^\mu \propto X^\mu$.
3. One is timelike (or null) and the other spacelike (or null), oriented so the spatial overlap cancels the temporal product — i.e. a genuinely orthogonal Minkowski pair.

A timelike 4-vector cannot be orthogonal to another timelike 4-vector (their inner product retains the sign of $c^2 \Delta t^2$).

Pairs among A, K, J .

Start from the normalization $U^\mu U_\mu = c^2$:

$$U^\mu U_\mu = c^2.$$

Differentiate with respect to τ :

$$2 A^\mu U_\mu = 0.$$

Conclude:

$$A^\mu U_\mu = 0.$$

Write J^μ for a particle of rest charge density ρ_0 :

$$J^\mu = \rho_0 U^\mu.$$

Contract with A_μ :

$$A^\mu J_\mu = \rho_0 A^\mu U_\mu = 0.$$

$A \cdot J = 0 \text{ always.}$

For K (null) and A (spacelike): no identity forces $A \cdot K = 0$; generically $A^\mu K_\mu \neq 0$. Likewise $K \cdot J = \rho_0 K^\mu U_\mu \propto \omega_{\text{rest}} \neq 0$ in general (the photon frequency seen by the charge). So only $A \cdot J = 0$ is guaranteed.

(b) 4-momentum, 4-acceleration, parallel case

Problem. Define $\mathbf{P}, \mathbf{A}$. Compute $P^\mu P_\mu$. Show $A^\mu A_\mu = \gamma^6 a^2$ when $\mathbf{v} \parallel \mathbf{a}$.

Define 4-momentum:

$$P^\mu = mU^\mu = (\gamma mc, \gamma m\mathbf{v}).$$

Define 4-acceleration:

$$A^\mu = \frac{dU^\mu}{d\tau} = \gamma \frac{dU^\mu}{dt}.$$

Form the Minkowski-square of P^μ :

$$P^\mu P_\mu = (\gamma mc)^2 - (\gamma m\mathbf{v}) \cdot (\gamma m\mathbf{v}).$$

Expand:

$$P^\mu P_\mu = \gamma^2 m^2 c^2 - \gamma^2 m^2 v^2.$$

Factor $\gamma^2 m^2$:

$$P^\mu P_\mu = \gamma^2 m^2 (c^2 - v^2).$$

Pull out c^2 :

$$P^\mu P_\mu = \gamma^2 m^2 c^2 (1 - v^2/c^2).$$

Apply $\gamma^2(1 - v^2/c^2) = 1$:

$$\boxed{P^\mu P_\mu = m^2 c^2.}$$

4-acceleration with $\mathbf{v} \parallel \mathbf{a}$. Write 4-velocity components:

$$U^0 = \gamma c, \quad \mathbf{U} = \gamma \mathbf{v}.$$

Differentiate the time component with respect to t :

$$\frac{dU^0}{dt} = c \dot{\gamma}.$$

Differentiate the spatial component:

$$\frac{d\mathbf{U}}{dt} = \dot{\gamma} \mathbf{v} + \gamma \mathbf{a}.$$

Differentiate $\gamma = (1 - v^2/c^2)^{-1/2}$ via the chain rule:

$$\dot{\gamma} = -\frac{1}{2}(1 - v^2/c^2)^{-3/2}(-2\mathbf{v} \cdot \mathbf{a}/c^2).$$

Simplify:

$$\dot{\gamma} = \gamma^3 \frac{\mathbf{v} \cdot \mathbf{a}}{c^2}.$$

Apply $\mathbf{v} \parallel \mathbf{a} \Rightarrow \mathbf{v} \cdot \mathbf{a} = va$:

$$\dot{\gamma} = \gamma^3 \frac{va}{c^2}.$$

Apply $A^\mu = \gamma dU^\mu/dt$ to the time component:

$$A^0 = \gamma \dot{\gamma} c.$$

Substitute $\dot{\gamma}$:

$$A^0 = \gamma^4 \frac{va}{c}.$$

Apply $A^\mu = \gamma dU^\mu/dt$ to the spatial component:

$$\mathbf{A} = \gamma(\dot{\gamma} \mathbf{v} + \gamma \mathbf{a}).$$

Substitute $\dot{\gamma}$ and $\mathbf{v} = v\hat{\mathbf{a}}$:

$$\mathbf{A} = \gamma^4 \frac{v^2 a}{c^2} \hat{\mathbf{a}} + \gamma^2 a \hat{\mathbf{a}}.$$

Factor $\gamma^2 a \hat{\mathbf{a}}$:

$$\mathbf{A} = \gamma^2 a \left(\gamma^2 \frac{v^2}{c^2} + 1 \right) \hat{\mathbf{a}}.$$

Apply $1 + \gamma^2 v^2/c^2 = \gamma^2$:

$$\mathbf{A} = \gamma^4 a \hat{\mathbf{a}}.$$

Take the magnitude:

$$|\mathbf{A}| = \gamma^4 a.$$

Form the Minkowski-square:

$$A^\mu A_\mu = (A^0)^2 - |\mathbf{A}|^2.$$

Substitute:

$$A^\mu A_\mu = \gamma^8 \frac{v^2 a^2}{c^2} - \gamma^8 a^2.$$

Factor $\gamma^8 a^2$:

$$A^\mu A_\mu = \gamma^8 a^2 \left(\frac{v^2}{c^2} - 1 \right).$$

Apply $v^2/c^2 - 1 = -1/\gamma^2$:

$$A^\mu A_\mu = -\gamma^8 a^2 / \gamma^2.$$

Simplify:

$$A^\mu A_\mu = -\gamma^6 a^2.$$

In signature $(+, -, -, -)$ the spacelike 4-acceleration has negative invariant; the question's $\gamma^6 a^2$ is the magnitude (equivalent to $(-, +, +, +)$ convention):

$$\boxed{|A^\mu A_\mu| = \gamma^6 a^2.}$$

(c) Inverse-square central force: energy conservation

Problem. For $\mathbf{f} = -\alpha \mathbf{r}/r^3$ (attractive, $\alpha > 0$), derive $\gamma m c^2 - \alpha/r = \text{const.}$

Start with the relativistic 2nd law (rest mass preserved):

$$\frac{d}{dt}(\gamma m \mathbf{v}) = \mathbf{f}.$$

Dot with $\mathbf{v}$:

$$\mathbf{v} \cdot \frac{d}{dt}(\gamma m \mathbf{v}) = \mathbf{f} \cdot \mathbf{v}.$$

Recognise the relativistic energy equation:

$$\frac{d}{dt}(\gamma m c^2) = \mathbf{f} \cdot \mathbf{v}.$$

Insert the attractive inverse-square force $\mathbf{f} = -\alpha \mathbf{r}/r^3$:

$$\mathbf{f} \cdot \mathbf{v} = -\frac{\alpha}{r^3} \mathbf{r} \cdot \mathbf{v}.$$

Use $\mathbf{r} \cdot \mathbf{v} = r\dot{r}$:

$$\mathbf{f} \cdot \mathbf{v} = -\frac{\alpha \dot{r}}{r^2}.$$

Use $\frac{d}{dt} = -\dot{r}/r^2$:

$$-\frac{\alpha \dot{r}}{r^2} = \alpha \frac{d}{dt} \left(\frac{1}{r} \right) = \frac{d}{dt} \left(\frac{\alpha}{r} \right).$$

Combine:

$$\frac{d}{dt}(\gamma m c^2) = \frac{d}{dt} \left(\frac{\alpha}{r} \right).$$

Collect into a single time derivative:

$$\frac{d}{dt} \left(\gamma m c^2 - \frac{\alpha}{r} \right) = 0.$$

$$\boxed{\gamma m c^2 - \frac{\alpha}{r} = \text{const.}}$$

Example. Electron-proton Coulomb attraction with $\alpha = e^2/(4\pi\epsilon_0)$, or Newtonian gravity with $\alpha = GMm$.

(d) **Electron accelerated across a 5 m gap**, $E = 10 \text{ MV/m}$

Problem. Find γ at exit, time of flight, sketch $\gamma(t), \beta(t), \Delta\beta$ if field reduced 20%, and proton case.
Apply the work-energy theorem:

$$\gamma mc^2 = mc^2 + eEL.$$

Solve for γ :

$$\gamma = 1 + \frac{eEL}{mc^2}.$$

Compute eEL with $E = 10 \text{ MV/m}$, $L = 5 \text{ m}$:

$$eEL = 10 \text{ MeV/m} \times 5 \text{ m} = 50 \text{ MeV}.$$

Insert $mc^2 = 0.511 \text{ MeV}$:

$$\gamma = 1 + \frac{50}{0.511}.$$

Evaluate the ratio:

$$\gamma = 1 + 97.85.$$

$$\boxed{\gamma \approx 98.85}$$

Time of flight. Write the 1-D equation of motion (constant force):

$$\frac{d}{dt}(\gamma mv) = eE.$$

Integrate from rest:

$$\gamma mv = eEt.$$

Square and apply $\gamma^2 v^2 = c^2(\gamma^2 - 1)$:

$$m^2 c^2 (\gamma^2 - 1) = (eEt)^2.$$

Solve for $\gamma(t)$:

$$\gamma(t) = \sqrt{1 + \left(\frac{eEt}{mc}\right)^2}.$$

Integrate $dx/dt = v = c\sqrt{1 - 1/\gamma^2}$:

$$x(t) = \frac{mc^2}{eE} \left[\sqrt{1 + (eEt/mc)^2} - 1 \right].$$

Set $x = L$:

$$\sqrt{1 + (eEt/mc)^2} = 1 + \frac{eEL}{mc^2} = \gamma.$$

Square:

$$\left(\frac{eEt}{mc}\right)^2 = \gamma^2 - 1.$$

Solve for t :

$$t = \frac{mc}{eE} \sqrt{\gamma^2 - 1}.$$

Compute $\sqrt{\gamma^2 - 1}$:

$$\sqrt{\gamma^2 - 1} = \sqrt{98.85^2 - 1}.$$

Evaluate:

$$\sqrt{\gamma^2 - 1} = \sqrt{9770.32} \approx 98.84.$$

Compute $mc^2/(eE)$:

$$\frac{mc^2}{eE} = \frac{0.511 \text{ MeV}}{10 \text{ MV/m}}.$$

Reduce the units:

$$\frac{mc^2}{eE} = 0.0511 \text{ m}.$$

Divide by c :

$$\frac{mc}{eE} = \frac{0.0511 \text{ m}}{c}.$$

Substitute into $t = (mc/eE)\sqrt{\gamma^2 - 1}$:

$$t = \frac{0.0511 \text{ m}}{c} \times 98.84.$$

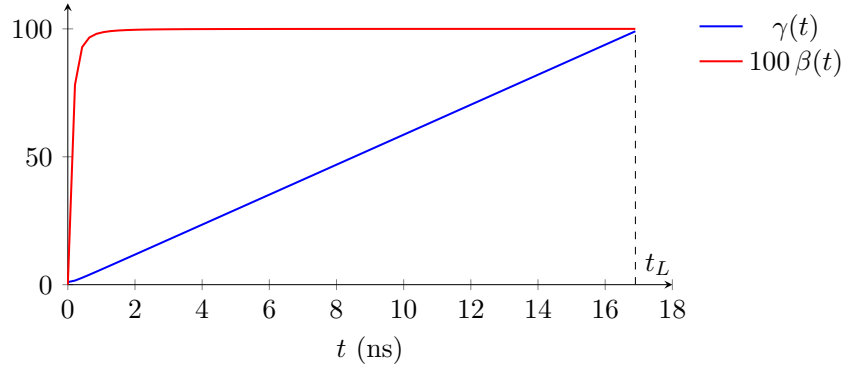
Multiply:

$$t = \frac{5.05 \text{ m}}{c}.$$

Insert $c = 2.998 \times 10^8 \text{ m/s}$:

$$t = \frac{5.05}{2.998 \times 10^8} \text{ s}.$$

$t \approx 1.69 \times 10^{-8} \text{ s} \approx 16.9 \text{ ns}.$



Field reduced by 20% ($E \rightarrow 0.8E$). Recompute γ' :

$$\gamma' = 1 + \frac{e(0.8E)L}{m_e c^2}.$$

Substitute $eEL/(m_e c^2) = 97.85$:

$$\gamma' = 1 + 0.8 \times 97.85 = 1 + 78.28.$$

Evaluate:

$$\gamma' = 79.28.$$

Both β, β' are very close to 1; expand $\beta = \sqrt{1 - 1/\gamma^2} \simeq 1 - 1/(2\gamma^2)$:

$$1 - \beta \simeq \frac{1}{2\gamma^2}.$$

Substitute $\gamma = 98.85$:

$$1 - \beta = \frac{1}{2 \times 98.85^2} = \frac{1}{19543}.$$

Evaluate:

$$1 - \beta \approx 5.12 \times 10^{-5}.$$

Substitute $\gamma' = 79.28$:

$$1 - \beta' = \frac{1}{2 \times 79.28^2} = \frac{1}{12571}.$$

Evaluate:

$$1 - \beta' \approx 7.96 \times 10^{-5}.$$

Form the difference using $\beta' - \beta = (1 - \beta) - (1 - \beta')$:

$$\Delta\beta = 5.12 \times 10^{-5} - 7.96 \times 10^{-5}.$$

$$\Delta\beta \approx -2.8 \times 10^{-5} \text{ (decrease).}$$

Proton instead. With $m_p c^2 = 938.3 \text{ MeV} \gg 50 \text{ MeV}$, motion is mildly relativistic. Compute γ_p :

$$\gamma_p = 1 + \frac{eEL}{m_p c^2} = 1 + \frac{50}{938.3}.$$

Evaluate the ratio:

$$\gamma_p = 1 + 0.0533 = 1.0533.$$

Compute γ_p^2 :

$$\gamma_p^2 = 1.0533^2 = 1.1094.$$

Form $1/\gamma_p^2$:

$$1/\gamma_p^2 = 1/1.1094 = 0.9014.$$

Apply $\beta_p = \sqrt{1 - 1/\gamma_p^2}$:

$$\beta_p = \sqrt{1 - 0.9014} = \sqrt{0.0986}.$$

Evaluate:

$$\beta_p = 0.314.$$

$$\gamma_p \approx 1.05, \beta_p \approx 0.31.$$

Hence the proton is far less relativistic ($\gamma_p \ll \gamma_e$, $\beta_p \ll 1$).

Question 2 — Photons, decays, fields, mirror reflection

(a) Photon 4-wavevector, phase invariance

Problem. Components of $\mathbf{K}$, dispersion, v_{ph} and v_g , show ϕ is Lorentz-invariant; is v_{ph} invariant?

Write the 4-wavevector:

$$K^\mu = (\omega/c, \mathbf{k}).$$

Apply vacuum dispersion:

$$\omega = c|\mathbf{k}|.$$

Form the null condition:

$$K^\mu K_\mu = \omega^2/c^2 - |\mathbf{k}|^2 = 0.$$

Define the phase velocity along $\hat{\mathbf{k}}$:

$$v_{\text{ph}} = \omega/|\mathbf{k}| = c.$$

Define the group velocity:

$$\mathbf{v}_g = \nabla_{\mathbf{k}}\omega = c\hat{\mathbf{k}}, \quad |\mathbf{v}_g| = c.$$

Write the phase of a plane wave:

$$\phi(\mathbf{r}, t) = \mathbf{k} \cdot \mathbf{r} - \omega t.$$

Identify the 4-vector contraction:

$$\phi = -K^\mu X_\mu.$$

A contraction of two 4-vectors is a Lorentz scalar:

$$\phi = -K^\mu X_\mu \text{ is Lorentz-invariant.}$$

Is v_{ph} Lorentz-invariant? For a vacuum photon, $\omega'/|\mathbf{k}'| = c$ in every frame, so yes — but only because c is the universal speed limit. In a medium with $v_{\text{ph}} \neq c$, a Lorentz boost generally changes both ω and $|\mathbf{k}|$ such that v_{ph} is *not* invariant. So:

$$v_{\text{ph}} = c \text{ in vacuum: invariant. In a medium: not invariant.}$$

(b) Forbidden and allowed photon processes

Problem. Can $\gamma \rightarrow m + \gamma'$? Photon-photon $\rightarrow e^+e^-$: minimum number, threshold energy.

Decay $\gamma \rightarrow m + \gamma'$. Write 4-momentum conservation using $\hbar K$ for photon 4-momenta:

$$\hbar K_1^\mu = P_m^\mu + \hbar K_2^\mu.$$

Isolate P_m^μ :

$$P_m^\mu = \hbar K_1^\mu - \hbar K_2^\mu.$$

Square both sides:

$$P_m^\mu P_{m\mu} = \hbar^2 K_1^\mu K_{1\mu} - 2\hbar^2 K_1^\mu K_{2\mu} + \hbar^2 K_2^\mu K_{2\mu}.$$

Apply $K_1^\mu K_{1\mu} = K_2^\mu K_{2\mu} = 0$ and $P_m^\mu P_{m\mu} = m^2 c^2$:

$$m^2 c^2 = -2\hbar^2 K_1^\mu K_{2\mu}.$$

Evaluate $K_1 \cdot K_2$ with θ_{12} the angle between the photons:

$$K_1^\mu K_{2\mu} = \omega_1 \omega_2 / c^2 - \mathbf{k}_1 \cdot \mathbf{k}_2.$$

Substitute $\mathbf{k}_1 \cdot \mathbf{k}_2 = (\omega_1 \omega_2 / c^2) \cos \theta_{12}$:

$$K_1^\mu K_{2\mu} = (\omega_1 \omega_2 / c^2)(1 - \cos \theta_{12}).$$

Insert into the squared equation:

$$m^2 c^2 = -2\hbar^2 (\omega_1 \omega_2 / c^2)(1 - \cos \theta_{12}).$$

Observe: LHS ≥ 0 , RHS ≤ 0 since $1 - \cos \theta_{12} \geq 0$ and $\omega_i > 0$. Equality requires $m = 0$ and $\theta_{12} = 0$ (trivial).

$$\text{Forbidden: a single photon cannot decay into } m + \gamma.$$

Photon-photon $\rightarrow e^+e^-$. A single photon has $K^\mu K_\mu = 0$ and cannot create a massive system ($P_{\text{pair}}^2 = (2m_e c)^2 > 0$) by itself. Two photons suffice if they are not collinear:

$$\hbar^2 (K_1 + K_2)^\mu (K_1 + K_2)_\mu = (2m_e c)^2.$$

Minimum: 2 photons.

Threshold. Equal frequencies, head-on collision: $\mathbf{k}_2 = -\mathbf{k}_1$ with $|\mathbf{k}_1| = |\mathbf{k}_2| = \omega/c$. Compute $K_1 \cdot K_2$:

$$K_1^\mu K_{2\mu} = \omega^2/c^2 - \mathbf{k}_1 \cdot \mathbf{k}_2.$$

Insert $\mathbf{k}_1 \cdot \mathbf{k}_2 = -\omega^2/c^2$:

$$K_1^\mu K_{2\mu} = \omega^2/c^2 + \omega^2/c^2 = 2\omega^2/c^2.$$

Expand $(K_1 + K_2)^2$:

$$(K_1 + K_2)^\mu (K_1 + K_2)_\mu = 2K_1^\mu K_{2\mu} = 4\omega^2/c^2.$$

Threshold (e^+e^- produced at rest in CM):

$$\hbar^2 \cdot 4\omega^2/c^2 = (2m_e c)^2.$$

Solve for ω :

$$\hbar\omega = m_e c^2.$$

Photon energy at threshold:

$$E_\gamma = \hbar\omega = m_e c^2.$$

$$E_\gamma^{\text{min}} = m_e c^2 \approx 0.511 \text{ MeV per photon.}$$

(c) Conserved 4-momentum components in $\mathbf{E}$ or $\mathbf{B}$ field

Problem. Electron with $\mathbf{v}_0 = (v_0, 0, 0)$. Which P^μ are conserved for (i) $\mathbf{E} = (E_x, 0, 0)$, (ii) $\mathbf{B} = (0, B_y, 0)$.

Lorentz force on electron (charge $q = -e$):

$$\frac{d\mathbf{p}}{dt} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Energy rate:

$$\frac{dE_{\text{tot}}}{dt} = -e\mathbf{E} \cdot \mathbf{v}.$$

(i) $\mathbf{E} = (E_x, 0, 0)$, **no** $\mathbf{B}$. Spatial force components:

$$\frac{dp_x}{dt} = -eE_x, \quad \frac{dp_y}{dt} = 0, \quad \frac{dp_z}{dt} = 0.$$

Energy rate evaluated:

$$\frac{dE_{\text{tot}}}{dt} = -eE_x v_x \neq 0.$$

Identify the conserved components:

$$P^y, P^z \text{ conserved; } P^0, P^x \text{ not.}$$

Initial conditions give $p_y = p_z = 0$ throughout.

(ii) $\mathbf{B} = (0, B_y, 0)$, **no** $\mathbf{E}$. Magnetic force does no work:

$$\frac{dE_{\text{tot}}}{dt} = 0 \Rightarrow P^0 \text{ conserved.}$$

Check the spatial force has no y -component. Compute $\mathbf{v} \times \mathbf{B}$ for general $\mathbf{v} = (v_x, v_y, v_z)$:

$$\mathbf{v} \times \mathbf{B} = (v_x, v_y, v_z) \times (0, B_y, 0) = (-v_z B_y, 0, v_x B_y).$$

The y -component vanishes identically, so:

$$\frac{dp_y}{dt} = 0.$$

With $p_y(0) = 0$, p_y remains zero. P^x and P^z rotate into each other (circular motion in xz -plane):

$P^0 \text{ and } P^y \text{ conserved; } P^x, P^z \text{ rotate (circular motion).}$

(d) Reflection from a relativistic mirror

Problem. Mirror moves at $\beta_x = 0.99$ along $\hat{\mathbf{x}}$. Lab photon $\lambda_1 = 1 \mu\text{m}$. In mirror frame $\mathbf{k}' = (-k'_x, k'_y, 0)$ with $|k'_x| = |k'_y|$. Find lab reflection angle and reflected λ_2 .

Step 1: Lorentz factor. Compute $1 - \beta^2$:

$$1 - \beta^2 = 1 - 0.99^2 = 1 - 0.9801 = 0.0199.$$

Take inverse square root:

$$\gamma = 1/\sqrt{0.0199} = 7.0888.$$

Step 2: Lab incident $\mathbf{K}_1$. In mirror frame, $|k'_x| = |k'_y|$ with $k'_x < 0$ (photon moves in $-\hat{\mathbf{x}}$ direction in the mirror frame). Write the 4-wavevector:

$$K_1'^{\mu} = (\omega'/c) (1, -1/\sqrt{2}, 1/\sqrt{2}, 0), \quad \omega' = c|k'_x|\sqrt{2}.$$

Boost from $S' \rightarrow S$ uses relative velocity $+v$ (the mirror frame moves at $+v\hat{\mathbf{x}}$ relative to lab):

$$\omega_1/c = \gamma(\omega'/c + \beta k'_{1x}).$$

Substitute $k'_{1x} = -(\omega'/c)/\sqrt{2}$:

$$\omega_1/c = \gamma(\omega'/c)(1 - \beta/\sqrt{2}).$$

Boost k_x component similarly:

$$k_{1x} = \gamma(k'_{1x} + \beta\omega'/c) = \gamma(\omega'/c)(\beta - 1/\sqrt{2}).$$

Transverse components unchanged:

$$k_{1y} = k'_{1y} = (\omega'/c)/\sqrt{2}.$$

Compute $\beta/\sqrt{2}$ with $\beta = 0.99$:

$$\beta/\sqrt{2} = 0.99/1.4142 = 0.7000.$$

Insert into ω_1/c :

$$\omega_1/c = 7.0888 (\omega'/c)(1 - 0.7000).$$

Multiply through:

$$\omega_1/c = 7.0888 \times 0.3000 (\omega'/c) = 2.127 \omega'/c.$$

Invert to express ω' in terms of given ω_1 :

$$\omega' = \omega_1/2.127.$$

Step 3: Reflection in mirror frame. A perfect mirror flips the normal component of $\mathbf{k}'$:

$$K_2'^{\mu} = (\omega'/c) (1, +1/\sqrt{2}, 1/\sqrt{2}, 0).$$

Reflection angle in mirror frame is 45° from normal.

Step 4: Boost reflected photon back to lab. Apply $S' \rightarrow S$ Lorentz transform:

$$\omega_2/c = \gamma(\omega'/c + \beta k_{2x}').$$

Substitute $k_{2x}' = +(\omega'/c)/\sqrt{2}$:

$$\omega_2/c = \gamma(\omega'/c)(1 + \beta/\sqrt{2}).$$

Boost the spatial component:

$$k_{2x} = \gamma(k_{2x}' + \beta\omega'/c) = \gamma(\omega'/c)(1/\sqrt{2} + \beta).$$

Transverse component unchanged:

$$k_{2y} = k_{2y}' = (\omega'/c)/\sqrt{2}.$$

Insert $\beta/\sqrt{2} = 0.7000$ into ω_2/c :

$$\omega_2/c = 7.0888 (\omega'/c)(1 + 0.7000).$$

Multiply through:

$$\omega_2/c = 7.0888 \times 1.7000 (\omega'/c) = 12.051 \omega'/c.$$

Form the frequency ratio:

$$\omega_2/\omega_1 = 12.051/2.127 = 5.666.$$

Use $\lambda \propto 1/\omega$:

$$\lambda_2 = \lambda_1 \omega_1/\omega_2 = 1 \mu\text{m}/5.666.$$

$$\boxed{\lambda_2 \approx 0.177 \mu\text{m} \approx 177 \text{ nm.}}$$

Reflection angle in lab. Measure from $+\hat{x}$ (mirror normal direction):

$$\tan \theta_2 = k_{2y}/k_{2x} = \frac{(\omega'/c)/\sqrt{2}}{\gamma(\omega'/c)(1/\sqrt{2} + \beta)}.$$

Cancel ω'/c :

$$\tan \theta_2 = \frac{1/\sqrt{2}}{\gamma(1/\sqrt{2} + \beta)} = \frac{0.7071}{7.0888 \times 1.6971}.$$

Compute denominator:

$$7.0888 \times 1.6971 = 12.030.$$

Evaluate:

$$\tan \theta_2 = 0.7071/12.030 = 0.05878.$$

$$\boxed{\theta_2 = \arctan(0.0588) \approx 3.36^\circ}$$

The reflected photon emerges almost along $+\hat{x}$ (relativistic beaming towards mirror motion direction).

Question 3 — Bounds, Compton, periodic-field undulator

(a) Bounds on forces, v_{ph} ; Compton shift

Problem. Bounds on forces/accelerations? On phase velocity? Derive Compton shift.

No upper bound on force or acceleration. Special relativity bounds the *velocity* of a massive particle ($v < c$); $\mathbf{a} = d\mathbf{v}/dt$ can be arbitrarily large transiently (only the integrated speed-gain is constrained). Similarly $\mathbf{f} = d\mathbf{p}/dt$ is unbounded since $|\mathbf{p}| = \gamma m |\mathbf{v}| \rightarrow \infty$ as $v \rightarrow c$.

Phase velocity is unbounded. In vacuum $v_{\text{ph}} = c$ exactly. In a medium $v_{\text{ph}} = \omega/|\mathbf{k}|$ can exceed c (e.g. X-rays in matter, plasmas above ω_p). Special relativity bounds only the group velocity / signal speed, not the phase velocity.

Compton shift. Write 4-momenta: photon $\hbar K = (\hbar\omega/c)(1, \hat{\mathbf{n}})$, electron at rest $P_e = (m_e c, \mathbf{0})$; after scatter $\hbar K' = (\hbar\omega'/c)(1, \hat{\mathbf{n}}')$, electron P'_e . 4-momentum conservation:

$$\hbar K + P_e = \hbar K' + P'_e.$$

Isolate P'_e :

$$P'_e = P_e + \hbar K - \hbar K'.$$

Square both sides:

$$P'^{\mu}_e P'_{e\mu} = (P_e + \hbar K - \hbar K')^{\mu} (P_e + \hbar K - \hbar K')_{\mu}.$$

Expand:

$$P'^2_e = P_e^2 + \hbar^2 K^2 + \hbar^2 K'^2 + 2\hbar P_e \cdot K - 2\hbar P_e \cdot K' - 2\hbar^2 K \cdot K'.$$

Apply $P_e^2 = P'^2_e = m_e^2 c^2$ and $K^2 = K'^2 = 0$:

$$m_e^2 c^2 = m_e^2 c^2 + 2\hbar P_e \cdot K - 2\hbar P_e \cdot K' - 2\hbar^2 K \cdot K'.$$

Cancel $m_e^2 c^2$ and divide by $2\hbar$:

$$P_e \cdot (K - K') = \hbar K \cdot K'.$$

Evaluate in rest frame $P_e = (m_e c, \mathbf{0})$:

$$P_e \cdot K = m_e c \cdot \omega / c = m_e \omega.$$

Similarly:

$$P_e \cdot K' = m_e \omega'.$$

Compute $K \cdot K'$ with $\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}' = \cos \theta$:

$$K \cdot K' = \omega \omega' / c^2 - (\omega \omega' / c^2) \cos \theta.$$

Factor:

$$K \cdot K' = (\omega \omega' / c^2) (1 - \cos \theta).$$

Substitute into the conservation identity:

$$m_e (\omega - \omega') = \hbar (\omega \omega' / c^2) (1 - \cos \theta).$$

Divide by $m_e \omega \omega'$:

$$\frac{1}{\omega'} - \frac{1}{\omega} = \frac{\hbar (1 - \cos \theta)}{m_e c^2}.$$

Use $\lambda = 2\pi c / \omega$ so $1/\omega = \lambda / (2\pi c)$:

$$\frac{\lambda' - \lambda}{2\pi c} = \frac{\hbar (1 - \cos \theta)}{m_e c^2}.$$

Solve for $\lambda' - \lambda$:

$$\lambda' - \lambda = \frac{2\pi\hbar}{m_e c} (1 - \cos \theta).$$

Use $h = 2\pi\hbar$:

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta) = \lambda_C (1 - \cos \theta)$$

with $\lambda_C = h/(m_e c) \approx 2.43 \text{ pm}$.

(b) Distance between two photons in S'

Problem. Two photons separated by x_0 along $\hat{x}$ in S . Find separation in S' moving with $\mathbf{v} = (v_x, 0, 0)$.

Take both photons propagating along $+\hat{x}$. In S at $t = 0$ the two emission/observation events are

$$E_A = (0, 0, 0, 0), \quad E_B = (0, x_0, 0, 0).$$

Apply the Lorentz boost along $+x$ at velocity v :

$$x' = \gamma(x - vt), \quad t' = \gamma(t - vx/c^2).$$

Transform E_A :

$$E'_A = (0, 0, 0, 0).$$

Transform E_B with $t = 0$, $x = x_0$:

$$x'_B = \gamma x_0, \\ t'_B = -\gamma v x_0 / c^2.$$

The events are not simultaneous in S' . Each photon's null worldline in S' is $x' = a_i + ct'$; solve for a_B :

$$a_B = x'_B - ct'_B.$$

Substitute:

$$a_B = \gamma x_0 - c(-\gamma v x_0 / c^2).$$

Simplify:

$$a_B = \gamma x_0 + \gamma v x_0 / c.$$

Factor γx_0 :

$$a_B = \gamma x_0 (1 + \beta).$$

Photon A passes through the origin, so $a_A = 0$. The separation at fixed t' is

$$\Delta x' = a_B - a_A = \gamma x_0 (1 + \beta).$$

Rewrite $\gamma = 1/\sqrt{(1 - \beta)(1 + \beta)}$:

$$\gamma(1 + \beta) = \frac{1 + \beta}{\sqrt{(1 - \beta)(1 + \beta)}} = \sqrt{\frac{1 + \beta}{1 - \beta}}.$$

$$\Delta x' = x_0 \gamma (1 + \beta) = x_0 \sqrt{\frac{1 + \beta}{1 - \beta}}.$$

This is the relativistic Doppler factor for a separation along the propagation direction (photons “stretched apart” when S' recedes).

(c) Head-on Compton: $\lambda_2 \approx \lambda_1/(4\gamma^2)$

Problem. Electron and photon counter-propagate; head-on backscatter. Show $\lambda_2 \approx \lambda_1/(4\gamma^2)$ when $\hbar\omega \ll m_e c^2$. Numerical for $\beta = 0.999$, $\lambda_1 = 8 \mu\text{m}$.

Method: Doppler shift twice. In the electron rest frame $\hbar\omega' \ll m_e c^2$ (hint), so the Compton shift is negligible and $\omega'_{\text{out}} \approx \omega'_{\text{in}} \equiv \omega'$.

Apply the relativistic Doppler formula lab $\rightarrow$ electron rest frame (head-on collision: photon along $-\hat{x}$, electron along $+\hat{x}$). Electron sees a blueshifted photon:

$$\omega' = \gamma(1 + \beta) \omega_1.$$

In the rest frame the photon backscatters: outgoing photon along $+\hat{x}$. Apply Doppler rest $\rightarrow$ lab (source moves $+\hat{x}$, observer at rest sees approaching source – blueshift):

$$\omega_2 = \gamma(1 + \beta) \omega'.$$

Combine the two boosts:

$$\omega_2 = \gamma^2(1 + \beta)^2 \omega_1.$$

Apply ultra-relativistic limit $\beta \rightarrow 1$:

$$(1 + \beta)^2 \rightarrow 4.$$

Hence:

$$\omega_2 \approx 4\gamma^2 \omega_1.$$

Use $\lambda \propto 1/\omega$:

$$\lambda_2 \approx \lambda_1/(4\gamma^2).$$

Numerical. Compute $1 - \beta^2$ for $\beta = 0.999$:

$$1 - \beta^2 = 1 - 0.998001 = 0.001999.$$

Take inverse square root:

$$\gamma = 1/\sqrt{0.001999} = 22.366.$$

Compute γ^2 :

$$\gamma^2 = 500.24.$$

Multiply by 4:

$$4\gamma^2 = 2000.95.$$

Form λ_2 :

$$\lambda_2 = 8 \mu\text{m}/2000.95.$$

Evaluate:

$$\lambda_2 \approx 4.0 \text{ nm}.$$

(d) Plane EM wave: A^μ , $\mathbf{E}$, $\mathbf{B}$

Problem. Plane wave ω , $\mathbf{k} = (k_x, 0, 0)$. Write a possible A^μ , find $\mathbf{E}$, $\mathbf{B}$.

Choose a Lorenz-gauge null 4-potential with $\hat{\mathbf{y}}$ polarization:

$$A^\mu = (0, 0, A_0 \cos(k_x x - \omega t), 0).$$

Apply $\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$ with $\phi = 0$:

$$\mathbf{E} = -\partial_t\mathbf{A}.$$

Differentiate A_y in time:

$$\partial_t A_y = A_0 \omega \sin(k_x x - \omega t).$$

Form $E_y = -\partial_t A_y$:

$$E_y = -A_0 \omega \sin(k_x x - \omega t).$$

Apply vacuum dispersion $\omega = ck_x$:

$$E_y = -A_0 ck_x \sin(k_x x - \omega t).$$

Apply $\mathbf{B} = \nabla \times \mathbf{A}$ with only $A_y(x, t)$ nonzero:

$$B_z = \partial_x A_y.$$

Differentiate:

$$B_z = -A_0 k_x \sin(k_x x - \omega t).$$

Collect:

$$\boxed{\mathbf{E} = -A_0 ck_x \sin(k_x x - \omega t) \hat{\mathbf{y}}, \quad \mathbf{B} = -A_0 k_x \sin(k_x x - \omega t) \hat{\mathbf{z}}.}$$

Check $|\mathbf{E}| = c|\mathbf{B}|$:

$$|\mathbf{E}|/|\mathbf{B}| = c.$$

Check the Poynting direction:

$$\mathbf{E} \times \mathbf{B} = E_y B_z (\hat{\mathbf{y}} \times \hat{\mathbf{z}}) = A_0^2 ck_x^2 \sin^2(k_x x - \omega t) \hat{\mathbf{x}}.$$

Transverse $\mathbf{E}, \mathbf{B}$ with $\mathbf{E} \times \mathbf{B} \parallel +\hat{\mathbf{x}}$ as required.

(e) Periodic-field undulator: lab vs. electron rest frame

Problem. Undulator $A^\mu = (\phi/c, \mathbf{0})$, $\phi = \phi_0 \cos(k_u z)$. Electron with $\mathbf{v} = (0, 0, v_z)$, ultra-relativistic. Find $\mathbf{E}, \mathbf{B}$ in lab and rest frame; is the field in the rest frame an EM wave?

Lab frame. With $A^0 = \phi/c$, $\mathbf{A} = \mathbf{0}$:

$$\mathbf{E} = -\nabla \phi - \partial_t \mathbf{A}.$$

Drop $\partial_t \mathbf{A}$:

$$\mathbf{E} = -\nabla \phi.$$

Differentiate $\phi = \phi_0 \cos(k_u z)$:

$$\nabla \phi = -\phi_0 k_u \sin(k_u z) \hat{\mathbf{z}}.$$

Negate:

$$\mathbf{E} = \phi_0 k_u \sin(k_u z) \hat{\mathbf{z}}.$$

With $\mathbf{A} = \mathbf{0}$:

$$\mathbf{B} = \nabla \times \mathbf{A} = \mathbf{0}.$$

$$\boxed{\mathbf{E}_{\text{lab}} = \phi_0 k_u \sin(k_u z) \hat{\mathbf{z}}, \quad \mathbf{B}_{\text{lab}} = \mathbf{0}.}$$

A purely electrostatic, longitudinal, z -periodic field.

Electron rest frame. Boost along $+\hat{\mathbf{z}}$ with velocity v . Coordinate inverse transform:

$$z = \gamma(z' + vt'), \quad t = \gamma(t' + vz'/c^2).$$

Apply the field transformation rule; $\mathbf{v} \parallel \mathbf{E}$, so the parallel component is unchanged:

$$E'_z = E_z.$$

Compute the perpendicular fields ($\mathbf{E}_\perp = 0$ and $\mathbf{B} = 0$):

$$\mathbf{E}'_\perp = \gamma(\mathbf{E}_\perp + \mathbf{v} \times \mathbf{B})_\perp = 0.$$

$$\mathbf{B}'_\perp = \gamma(\mathbf{B}_\perp - \mathbf{v} \times \mathbf{E}/c^2)_\perp.$$

Apply $\mathbf{v} \times \mathbf{E} = 0$ (since $\mathbf{v} \parallel \mathbf{E}$):

$$\mathbf{B}'_\perp = 0.$$

Substitute $z = \gamma(z' + vt')$ into the spatial argument:

$$k_u z = \gamma k_u z' + \gamma k_u v t'.$$

Identify $k'_u = \gamma k_u$ and $\omega' = \gamma k_u v$:

$$E'_z(\mathbf{r}', t') = \phi_0 k_u \sin(\gamma k_u z' + \gamma k_u v t').$$

$$\boxed{\mathbf{E}' = \phi_0 k_u \sin(\gamma k_u z' + \gamma k_u v t') \hat{\mathbf{z}}', \quad \mathbf{B}' = \mathbf{0}.}$$

In the ultra-relativistic limit $v \rightarrow c$: $\omega' \approx \gamma k_u c$ and $\omega'/k'_u = v \approx c$.

Is this an EM wave? A vacuum plane EM wave satisfies $|\mathbf{E}| = c|\mathbf{B}|$ with $\mathbf{E} \perp \mathbf{k}$ and $\mathbf{B} \perp \mathbf{k}$, $\mathbf{B} \neq 0$. Here $\mathbf{B}' = 0$ and $\mathbf{E}' \parallel \hat{\mathbf{z}}'$ (longitudinal to the propagation direction $\hat{\mathbf{z}}'$).

$$\boxed{\text{Not an EM wave: longitudinal } \mathbf{E}, \mathbf{B}' = 0.}$$

It is a longitudinal oscillating electric field — the rest-frame view of a static periodic potential as seen by a co-moving observer (no radiation; not transverse).

Question 4 — Polar/axial vectors, $F^{\mu\nu}$, capacitor

(a) Polar vs. axial; $\mathbf{E}, \mathbf{B}$

Problem. Define polar/axial 3-vectors. Are $\mathbf{E}, \mathbf{B}$ polar or axial? Justify via Lorentz force.

Polar vector: flips sign under spatial inversion $\mathbf{r} \rightarrow -\mathbf{r}$ (e.g. $\mathbf{r}, \mathbf{v}, \mathbf{p}, \mathbf{E}$).

Axial (pseudo-)vector: unchanged under inversion (e.g. $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, $\mathbf{B}, \boldsymbol{\omega}$).

Write the Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Apply parity $\mathbf{r} \rightarrow -\mathbf{r}$, $\mathbf{v} \rightarrow -\mathbf{v}$, scalar q unchanged. Since $\mathbf{F} \propto d\mathbf{p}/dt$ is polar:

$$\mathbf{F} \rightarrow -\mathbf{F}.$$

Write the transformed equation in primed fields:

$$-\mathbf{F} = q\mathbf{E}' + q(-\mathbf{v}) \times \mathbf{B}'.$$

Set equal to $-\mathbf{F} = -q\mathbf{E} - q\mathbf{v} \times \mathbf{B}$:

$$-q\mathbf{E} - q\mathbf{v} \times \mathbf{B} = q\mathbf{E}' - q\mathbf{v} \times \mathbf{B}'.$$

Match the $\mathbf{v}$ -independent term:

$$\mathbf{E}' = -\mathbf{E}.$$

Match the $\mathbf{v}$ -dependent term:

$$-q\mathbf{v} \times \mathbf{B} = -q\mathbf{v} \times \mathbf{B}'.$$

Solve:

$$\mathbf{B}' = \mathbf{B}.$$

Hence $\mathbf{E}$ flips sign (polar) and $\mathbf{B}$ does not (axial).

$\mathbf{E} \text{ polar, } \mathbf{B} \text{ axial.}$

(b) 4-current, continuity, Lorentz force on a charge density

Problem. Define J^μ , write continuity, derive $f_\mu = J^\nu F_{\nu\mu}$, identify f_0 .

Define 4-current:

$$J^\mu = \rho_0 U^\mu = (\rho c, \mathbf{j}), \quad \mathbf{j} = \rho \mathbf{v}.$$

Write continuity:

$\partial_\mu J^\mu = 0 \quad \Leftrightarrow \quad \partial_t \rho + \nabla \cdot \mathbf{j} = 0.$

Derive $f_\mu = J^\nu F_{\nu\mu}$. Write the field tensor (covariant form):

$$F_{\nu\mu} = \partial_\nu A_\mu - \partial_\mu A_\nu.$$

List components in signature $(+, -, -, -)$:

$$F_{0i} = E_i/c, \quad F_{ij} = -\epsilon_{ijk} B_k.$$

Take $\mu = i$ in $J^\nu F_{\nu i}$:

$$J^\nu F_{\nu i} = J^0 F_{0i} + J^j F_{ji}.$$

Apply $F_{ji} = -F_{ij} = \epsilon_{ijk} B_k$:

$$J^\nu F_{\nu i} = (\rho c)(E_i/c) + j_j \epsilon_{ijk} B_k.$$

Recognise $\epsilon_{ijk} j_j B_k = (\mathbf{j} \times \mathbf{B})_i$:

$$J^\nu F_{\nu i} = \rho E_i + (\mathbf{j} \times \mathbf{B})_i.$$

Identify $f_i = \rho E_i + (\mathbf{j} \times \mathbf{B})_i$ as the Lorentz force density:

$$J^\nu F_{\nu i} = f_i.$$

f_0 **component:** Take $\mu = 0$:

$$J^\nu F_{\nu 0} = J^0 F_{00} + J^j F_{j0}.$$

Apply $F_{00} = 0$ and $F_{j0} = -F_{0j} = -E_j/c$:

$$J^\nu F_{\nu 0} = -j^j E_j/c.$$

Identify $j^j E_j = \mathbf{j} \cdot \mathbf{E}$:

$$|f_0| = \mathbf{j} \cdot \mathbf{E}/c.$$

Interpret: $|f_0| \cdot c = \mathbf{j} \cdot \mathbf{E}$ is the power per unit volume delivered by the field to the charges.

$|f_0| c = \mathbf{j} \cdot \mathbf{E} \text{ (power per unit volume delivered by field to charges).}$

(c) $F^{\alpha\beta}$ from given A^μ

Problem. $A^\mu = (0, A_0 \cos(K \cdot X), A_0 \sin(K \cdot X), 0)$, $K = (\omega/c, 0, 0, k_z)$. Compute $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$.

Define the phase:

$$\Phi \equiv K^\mu X_\mu = (\omega/c)(ct) + k_z(-z) = \omega t - k_z z.$$

List components: $A^1 = A_0 \cos \Phi$, $A^2 = A_0 \sin \Phi$, $A^0 = A^3 = 0$.

Differentiate the phase:

$$\partial_0 \Phi = \omega/c, \quad \partial_3 \Phi = -k_z.$$

Apply $\partial^0 = \partial_0$ and $\partial^i = -\partial_i$.

Compute $\partial^0 A^1$:

$$\partial^0 A^1 = (1/c) \partial_t (A_0 \cos \Phi) = -A_0 (\omega/c) \sin \Phi.$$

Compute $\partial^0 A^2$:

$$\partial^0 A^2 = (1/c) \partial_t (A_0 \sin \Phi) = A_0 (\omega/c) \cos \Phi.$$

Compute $\partial^3 A^1$:

$$\partial^3 A^1 = -\partial_z (A_0 \cos \Phi) = -A_0 \sin \Phi \cdot k_z = -A_0 k_z \sin \Phi.$$

Compute $\partial^3 A^2$:

$$\partial^3 A^2 = -\partial_z (A_0 \sin \Phi) = -A_0 \cos \Phi \cdot (-k_z) = A_0 k_z \cos \Phi.$$

All other $\partial^\mu A^\nu$ vanish (no x, y dependence).

Assemble $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$:

$$F^{01} = \partial^0 A^1 - \partial^1 A^0 = -A_0 (\omega/c) \sin \Phi.$$

$$F^{02} = \partial^0 A^2 - \partial^2 A^0 = A_0 (\omega/c) \cos \Phi.$$

$$F^{31} = \partial^3 A^1 - \partial^1 A^3 = -A_0 k_z \sin \Phi.$$

$$F^{32} = \partial^3 A^2 - \partial^2 A^3 = A_0 k_z \cos \Phi.$$

$$F^{12} = \partial^1 A^2 - \partial^2 A^1 = 0.$$

Identify $\mathbf{E}$ via $F^{0i} = -E_i/c$:

$$E_x = -c F^{01} = A_0 \omega \sin \Phi.$$

$$E_y = -c F^{02} = -A_0 \omega \cos \Phi.$$

$$E_z = 0.$$

Identify $\mathbf{B}$ via $F^{ij} = -\epsilon^{ijk} B_k$; so $F^{31} = -\epsilon^{312} B_2 = -B_y$ and $F^{32} = -\epsilon^{321} B_1 = +B_x$:

$$B_x = F^{32} = A_0 k_z \cos \Phi.$$

$$B_y = -F^{31} = A_0 k_z \sin \Phi.$$

$$B_z = 0.$$

Collect:

$$\mathbf{E} = (A_0 \omega \sin \Phi, -A_0 \omega \cos \Phi, 0).$$

$$\mathbf{B} = (A_0 k_z \cos \Phi, A_0 k_z \sin \Phi, 0).$$

Compute magnitudes and orthogonality:

$$|\mathbf{E}| = A_0 \omega, \quad |\mathbf{B}| = A_0 k_z, \quad |\mathbf{E}| = c |\mathbf{B}| \text{ (use } \omega = ck_z \text{)}.$$

$$\mathbf{E} \cdot \mathbf{B} = A_0^2 \omega k_z (\sin \Phi \cos \Phi - \cos \Phi \sin \Phi) = 0.$$

$$\mathbf{E} \perp \mathbf{B}, |\mathbf{E}| = c|\mathbf{B}|; \text{ circularly polarised plane wave along } +\hat{\mathbf{z}}.$$

Verify Poynting along $+\hat{\mathbf{z}}$:

$$(\mathbf{E} \times \mathbf{B})_z = E_x B_y - E_y B_x.$$

Substitute:

$$(\mathbf{E} \times \mathbf{B})_z = (A_0 \omega \sin \Phi)(A_0 k_z \sin \Phi) - (-A_0 \omega \cos \Phi)(A_0 k_z \cos \Phi).$$

Simplify:

$$(\mathbf{E} \times \mathbf{B})_z = A_0^2 \omega k_z (\sin^2 \Phi + \cos^2 \Phi) = A_0^2 \omega k_z.$$

(d) Field transformation under boost

Problem. S' moves with 3-velocity $\mathbf{v}$ relative to S . Write $\mathbf{E}'$, $\mathbf{B}'$.

Decompose into components parallel ($\parallel$) and perpendicular ($\perp$) to $\mathbf{v}$. Parallel components unchanged:

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad \mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}.$$

Perpendicular components transform with γ and mixing:

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}).$$

$$\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - (\mathbf{v} \times \mathbf{E})/c^2).$$

$$\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}), \quad \mathbf{B}' = \mathbf{B}_{\parallel} + \gamma(\mathbf{B}_{\perp} - (\mathbf{v} \times \mathbf{E})/c^2).$$

(e) Capacitor + magnetic field

Problem. Plates parallel to yz -plane, gap $d = 0.1$ m, $E = 100$ MV/m between plates, electron launched from negative plate at rest. (i) Energy and velocity at the second plate; (ii) can a $\mathbf{B}$ prevent it hitting positive plate; (iii) direction; (iv) magnitude; (v) trajectory sketch.

(i) Energy and velocity at the second plate.

Compute kinetic energy from work done by $\mathbf{E}$:

$$E_{\text{kin}} = eEd.$$

Substitute $E = 100$ MV/m, $d = 0.1$ m:

$$E_{\text{kin}} = e \cdot 100 \text{ MV/m} \cdot 0.1 \text{ m} = 10 \text{ MeV}.$$

Form the total energy:

$$E_{\text{tot}} = m_e c^2 + E_{\text{kin}} = 0.511 + 10 = 10.511 \text{ MeV}.$$

Compute the Lorentz factor:

$$\gamma = E_{\text{tot}}/(m_e c^2) = 10.511/0.511.$$

Evaluate:

$$\gamma = 20.57.$$

Compute $1/\gamma^2$:

$$1/\gamma^2 = 1/423.1 = 0.002364.$$

Apply $\beta = \sqrt{1 - 1/\gamma^2}$:

$$\beta = \sqrt{1 - 0.002364} = \sqrt{0.99764} = 0.99882.$$

$$E_{\text{kin}} \approx 10 \text{ MeV}, \quad \gamma \approx 20.6, \quad v \approx 0.999 c.$$

(ii) Can B prevent it reaching the positive plate?

Yes. A magnetic field perpendicular to the electron velocity bends its trajectory; if the field is strong enough, the gyroradius is smaller than the gap so that the electron curves back to the negative plate before hitting the positive plate.

Yes — a transverse B bends the electron back.

(iii) Direction of B . Take electron initial $\mathbf{v}$ along $+\hat{x}$. To bend it inside the xz (or xy) plane back toward the negative plate, B must lie in the yz -plane (perpendicular to $\mathbf{v}$). Choosing $B = B\hat{y}$:

$B \parallel \hat{y}$ (or $\hat{z}$): perpendicular to $\mathbf{v}$ and to $\mathbf{E}$.

The electron then executes (modified) cyclotron motion in the xz -plane.

(iv) Strength of B to stop the electron at $x = d$.

Energy is conserved by $\mathbf{E}$ alone (B does no work):

$$\gamma(x)m_e c^2 = m_e c^2 + eEx.$$

Choose gauge $\mathbf{A} = -Bx\hat{z}$ for $B = B\hat{y}$:

$$\nabla \times \mathbf{A} = -(\partial_x A_z)\hat{y} = B\hat{y} \quad \checkmark.$$

System has translation symmetry along z , so z -component of canonical momentum is conserved:

$$\Pi_z = p_z + qA_z = \text{const}, \quad q = -e.$$

At launch $p_z = 0$, $x = 0 \Rightarrow A_z = 0$:

$$\Pi_z = 0.$$

Substitute $A_z = -Bx$ and $q = -e$ at general x :

$$p_z(x) + (-e)(-Bx) = 0.$$

Solve for $p_z(x)$:

$$p_z(x) = -eBx.$$

At turn-around $v_x = 0$, so $p_x = 0$ and $|p_z| = p_{\text{tot}}$:

$$p_{\text{tot}}(x_{\text{max}}) = eBx_{\text{max}}.$$

Apply energy-momentum relation:

$$(p_{\text{tot}}c)^2 = E_{\text{tot}}^2 - (m_e c^2)^2.$$

Set turn-around at $x_{\text{max}} = d$:

$$eBd = p_{\text{tot}}(d).$$

Solve for B :

$$B = p_{\text{tot}}(d)/(ed).$$

Compute $(p_{\text{tot}}c)^2$ at $E_{\text{tot}} = 10.511 \text{ MeV}$:

$$(p_{\text{tot}}c)^2 = 10.511^2 - 0.511^2.$$

Evaluate:

$$(p_{\text{tot}}c)^2 = 110.481 - 0.261 = 110.22 (\text{MeV})^2.$$

Take the square root:

$$p_{\text{tot}}c = 10.499 \text{ MeV}.$$

Form $B = (p_{\text{tot}}c)/(ecd)$:

$$B = \frac{p_{\text{tot}}c}{ecd} = \frac{10.499 \times 10^6 \text{ V}}{c 0.1 \text{ m}}.$$

Insert $c = 2.998 \times 10^8 \text{ m/s}$:

$$B = \frac{10.499 \times 10^6 \text{ V}}{2.998 \times 10^8 \text{ m/s} \times 0.1 \text{ m}}.$$

Compute denominator:

$$(2.998 \times 10^8 \text{ m/s})(0.1 \text{ m}) = 2.998 \times 10^7 \text{ m}^2/\text{s}.$$

Divide:

$$B = \frac{10.499 \times 10^6 \text{ V}}{2.998 \times 10^7 \text{ m}^2/\text{s}} = 0.350 \text{ T}.$$

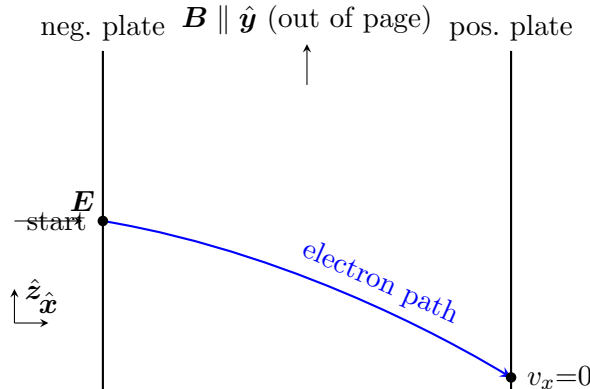
$$\boxed{B \approx 0.35 \text{ T}.$$

(v) Trajectory sketch.

The electron starts at the negative plate with $\mathbf{v} = \mathbf{0}$. Initially $\mathbf{B}$ exerts no force; the $\mathbf{E}$ -field accelerates it along $+\hat{x}$. As v_x grows the magnetic force becomes:

$$\mathbf{F}_{\text{mag}} = -e\mathbf{v} \times \mathbf{B} = -ev_x\hat{x} \times B\hat{y} = -ev_xB\hat{z}.$$

This deflects the electron toward $-\hat{z}$. The trajectory curves; at the threshold $B = 0.35 \text{ T}$, $v_x \rightarrow 0$ exactly at $x = d$ (the electron just grazes the positive plate before turning back). For B slightly above threshold, the electron turns back before reaching $x = d$ and executes a cycloidal drift in the xz -plane.



If B slightly exceeds threshold, the electron turns back before $x = d$ and executes a cycloidal drift along $-\hat{z}$ between the plates (drift velocity $\mathbf{v}_d = \mathbf{E} \times \mathbf{B}/B^2$).